TIES Communications, Inc.

Technical Capability Addendum

Engineering Depth & Specialized Network Expertise · For Prime Contractor Technical Evaluators

> This addendum supplements the TIES Communications Statement of Capabilities. It is intended for prime contractor engineering teams, technical evaluators, and solution architects who require deeper insight into how TIES designs, implements, and troubleshoots complex network and infrastructure environments supporting federal mission requirements.

Engineering-Led Delivery Model

TIES Communications operates under an engineering-first delivery philosophy. Network solutions are developed through formal analysis and modeling — not vendor defaults or prescriptive templates.

Design and remediation efforts incorporate:

- Protocol-level behavior analysis
- Mathematical performance modeling
- Empirical validation and controlled testing

This approach enables accurate diagnosis and resolution of systemic issues including:

- MTU fragmentation and blackholing
- Latency-driven throughput collapse
- Asymmetric routing and path instability
- Congestion and queue-management failures

> *Problems are modeled, validated, and corrected — not guessed.*

Core Technical Competencies

Enterprise & Mission-Critical Network Architecture

- LAN/WAN architectures supporting high availability, segmentation, and fault tolerance
- Zero-Trust-aligned designs incorporating least-privilege principles
- Multi-site and geographically distributed network environments

Carrier-Grade Routing & Switching

- BGP, OSPF, MPLS
- EVPN and overlay networking
- SD-WAN architectures (VeloCloud and platform-agnostic)
- Operational experience at the same protocol layer used by Tier 1 integrators and service providers

Hybrid Data Center & Cloud-Ready Design

- On-premises and hybrid data center networking
- Integration of legacy environments with modern architectures
- Design patterns that support gradual modernization without operational disruption

Wireless, RF & Specialized Communications

- Enterprise wireless deployments
- Point-to-point microwave and RF solutions
- VHF, UHF, and satellite-supporting frequency ranges
- Remote, expeditionary, and infrastructure-limited environments

Structured Cabling & Physical Plant

- Enterprise copper — Cat5e/Cat6 — deployment and certification

- Single-mode and multi-mode fiber
- Outside plant (OSP) infrastructure
- Remediation, certification, and lifecycle management

Performance Engineering & Advanced Troubleshooting

TIES Communications specializes in resolving complex, multi-layer performance issues — often escalated after standard troubleshooting methodologies have failed.

Common engagements include:

- Multi-vendor interoperability failures
- Application performance degradation traced to network behavior
- MTU propagation errors across hybrid or tunneled paths
- Traffic engineering and routing instability
- Latency and packet-loss analysis impacting voice, video, or mission systems

Solutions are validated through controlled testing, post-deployment monitoring, and measured performance baselines.

Network Automation & Analytics

Engineering teams employ automation and analytics to improve reliability and operational visibility:

- Infrastructure scripting using Python and R
- Telemetry collection and performance analysis
- Performance trend modeling and anomaly detection
- Configuration consistency and deployment automation

Automation reduces human error, improves repeatability, and supports scalable operations across multi-site deployments.

Tools, Platforms & Technical Ecosystem

TIES Communications maintains hands-on deployment and operational experience across enterprise and carrier-class platforms:

- HPE Networking — Aruba & Juniper
- HPE Compute
- Schneider Electric / APC — power and cooling solutions
- Veeam — backup and recovery
- Infoblox — DNS, DHCP, and IP address management

> *Architectural design remains vendor-agnostic — solutions align with Prime standards and government mandates, not reseller preferences.*

Compliance-Aligned Engineering

Technical implementations are designed to support federal compliance requirements:

- FCI protection under 48 CFR 52.204-21
- Current CMMC Level 1 practices with Level 2 preparation underway

Engineering decisions prioritize:

- Secure management planes and access control
- Network segmentation aligned to least-privilege principles
- Audit-friendly configurations and documentation
- Technical design that supports — rather than complicates — Prime compliance obligations

Engagement Model & Prime Support

TIES Communications most often supports Prime contractors as a specialized engineering subcontractor, providing:

- Design validation and peer review

- Deployment execution
- Modernization and legacy infrastructure support
- Escalation-level troubleshooting for complex performance issues

Engagements scale from focused technical problem-solving to full-lifecycle infrastructure modernization. The lean two-principal structure ensures direct access to senior engineering expertise on every engagement — no layered project management between the Prime and the engineer.

> This Technical Capability Addendum is intended to be used in conjunction with the TIES Communications Statement of Capabilities. Together, the documents provide executive clarity and compliance confidence alongside deep technical assurance for engineering evaluators.

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